DC-DC CONVERTER (i-ELOOP) [i-ELOOP]

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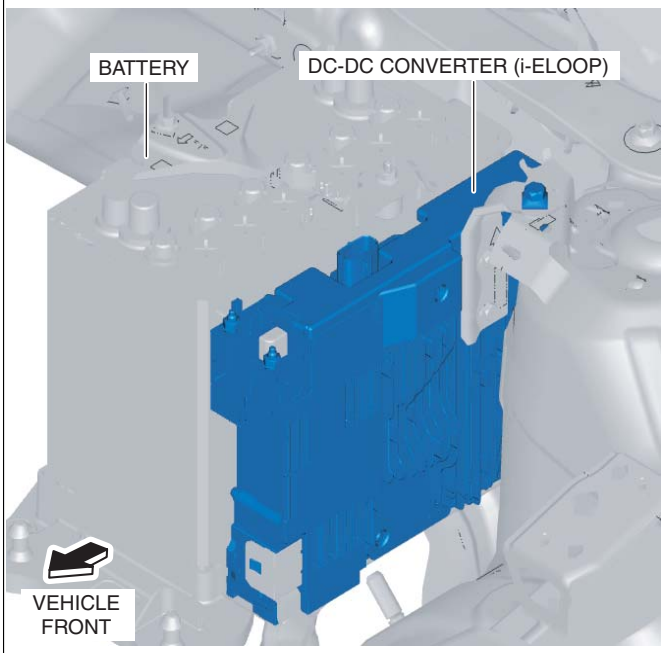
Purpose, Function

- To supply the power generated by the generator to the vehicle's electrical devices, the DC-DC converter (i-ELOOP) performs control according to the engine operation conditions.
- To prevent electrical circuit failure, part over-heating, or ignition due to large amounts of current, if the power supplied from the capacitor (i-ELOOP) exceeds the rated voltage of the vehicle's electrical devices, power cannot be supplied directly to the vehicle's electrical devices. As a result, the DC-DC converter (i-ELOOP) lowers the maximum 24.3 V voltage to 12 V, and then supplies power to the vehicle's electrical devices.

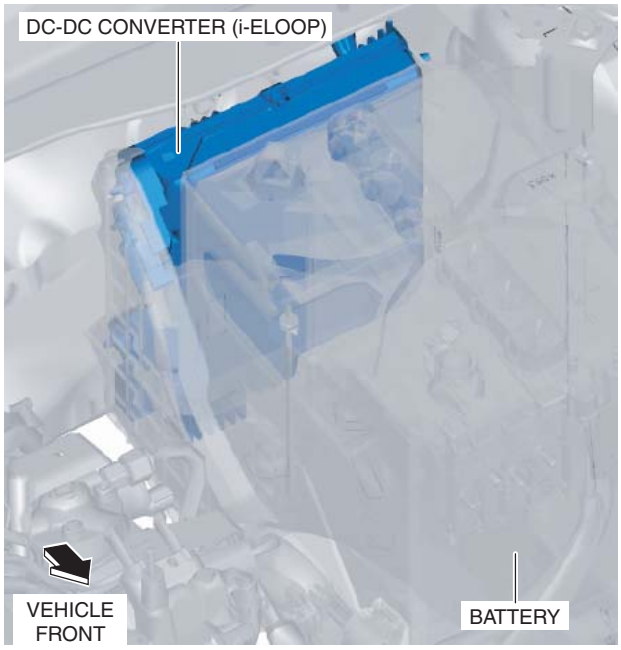
Construction

- The DC-DC converter (i-ELOOP) is located behind the battery.

SKYACTIV-G 1.5

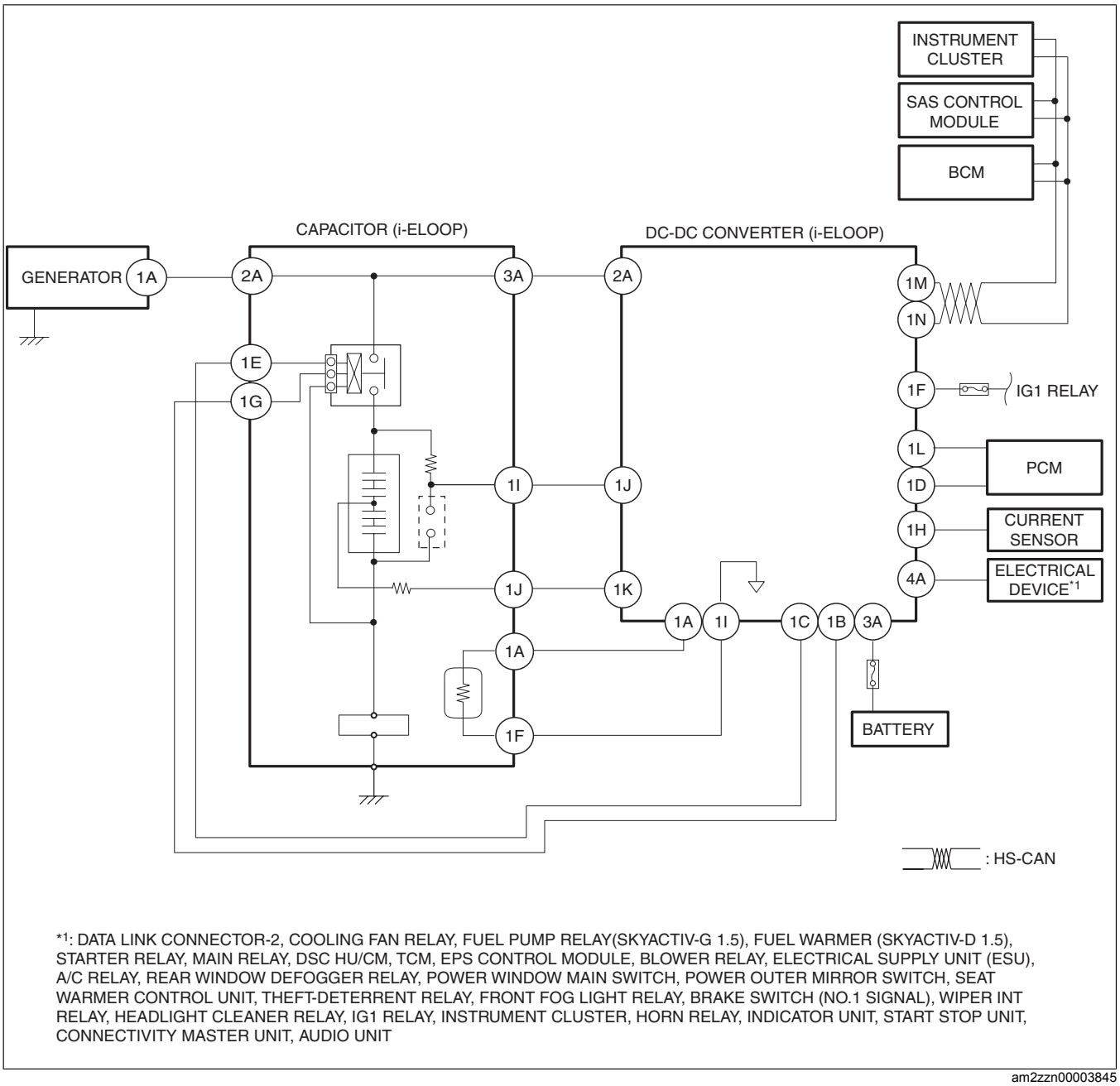


SKYACTIV-D 1.5



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- The DC-DC converter (i-ELOOP) has the following inputs/outputs.



*1: DATA LINK CONNECTOR-2, COOLING FAN RELAY, FUEL PUMP RELAY(SKYACTIV-G 1.5), FUEL WARMER (SKYACTIV-D 1.5), STARTER RELAY, MAIN RELAY, DSC HU/CM, TCM, EPS CONTROL MODULE, BLOWER RELAY, ELECTRICAL SUPPLY UNIT (ESU), A/C RELAY, REAR WINDOW DEFOGGER RELAY, POWER WINDOW MAIN SWITCH, POWER OUTER MIRROR SWITCH, SEAT WARMER CONTROL UNIT, THEFT-DETERRENT RELAY, FRONT FOG LIGHT RELAY, BRAKE SWITCH (NO.1 SIGNAL), WIPER INT RELAY, HEADLIGHT CLEANER RELAY, IG1 RELAY, INSTRUMENT CLUSTER, HORN RELAY, INDICATOR UNIT, START STOP UNIT, CONNECTIVITY MASTER UNIT, AUDIO UNIT

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- The DC-DC converter (i-ELOOP) switches the operation content according to the vehicle conditions. Refer to the [i-ELOOP] for details. (See i-ELOOP [i-ELOOP].)

Fail-safe

DTC	Fail-safe
P0A12:00	• Inhibits engine-stop by operating the i-stop function.
P2502:00	
P2503:00	
P2504:00	
P1794:00	• Inhibits engine-stop by operating the i-stop function. (With i-stop) • Inhibits the i-ELOOP control. • Switches to bypass mode which shuts the capacitor (i-ELOOP) off.
U0001:88	• Switches to bypass mode which shuts the capacitor (i-ELOOP) off.
U0100:00	
U3000:49	• Inhibits engine-stop by operating the i-stop function. (With i-stop) • Switches to bypass mode which shuts the capacitor (i-ELOOP) off.